

Chernoff bound

Lee-Ad Gottlieb

Chernoff bound

- Let **B** be a Bernoulli $\{0,1\}$ distribution with parameter p :
 - $\Pr[x = 1] = p$ $\Pr[x=0] = 1-p$
- Draw n random independently distributed trials $\sim$ **B**, and let $X = \frac{1}{n} \sum_i X_i$ be their empirical mean. Let $\epsilon \leq 1$ and note that $p = E[X]$
- Then $\Pr[X > (1 + \epsilon)p] < e^{-np\epsilon^2/3}$

Preliminaries

- 1. Markov: For any random variable $Y \geq 0$,

$$\Pr[Y > c] < \frac{E[Y]}{c} \qquad \Pr[Y > cE[Y]] < \frac{1}{c}$$

- 2. Taylor expansion at 0:

$$- f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

$$\begin{aligned} - \ln(1+x) &= 0 + x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots \\ &> x - \frac{x^2}{2} + \frac{x^3}{3} \end{aligned}$$

$$\begin{aligned} - \text{For } x < 1: (1+x) \ln(1+x) &> (1+x) \left(x - \frac{x^2}{2} + \frac{x^3}{3} \right) \\ &= x + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{3} \geq x + \frac{x^2}{3} \end{aligned}$$

Proof

- For any $t > 0$,

$$\begin{aligned}\Pr[X > (1 + \epsilon)p] &= \Pr[e^{tX} > e^{t(1+\epsilon)p}] \\ &\leq \frac{E[e^{tX}]}{e^{t(1+\epsilon)p}}\end{aligned}$$

Markov

- Now let's focus on $E[e^{tx}]$

Proof

- $E[e^{tX}] = E\left[e^{\left(\frac{t}{n}\right) \sum_i X_i}\right]$
- $= E\left[\prod_i e^{(t/n)X_i}\right]$
- $= \prod_i E\left[e^{(t/n)X_i}\right]$
- $= \prod_i \left(pe^{t/n} + (1-p)\right)$
- $= \prod_i \left(1 + p(e^{t/n} - 1)\right)$
- $\leq \prod_i e^{p(e^{t/n} - 1)}$
- $= e^{np(e^{t/n} - 1)}$
- $= e^{np(e^{t/n} - 1)}$

independence

definition of expectation,
with $p = \Pr[X_i = 1]$
and $1-p = \Pr[X_i = 0]$

$1+x \leq e^x$

Proof

- So for any $t > 0$,

$$\begin{aligned}\Pr[X > (1 + \epsilon)p] &= \Pr[e^{tX} > e^{t(1+\epsilon)p}] \\ &\leq \frac{E[e^{tx}]}{e^{t(1+\epsilon)p}} \\ &\leq \frac{e^{np(e^{t/n}-1)}}{e^{t(1+\epsilon)p}}\end{aligned}$$

- Take $t = n \ln(1 + \epsilon)$:

$$\begin{aligned}\Pr[X > (1 + \epsilon)p] &\leq e^{np[\epsilon - (1+\epsilon) \ln(1+\epsilon)]} \\ &\leq e^{-np(\frac{\epsilon^2}{3})}\end{aligned}$$

$$\text{Use } (1 + \epsilon) \ln(1 + \epsilon) > \epsilon + \frac{\epsilon^2}{3}$$